



Starter File: HeatXStarter.bkp
heatx_PlantData_historian.xlsm

Solution file: HeatXTrained.apwz

Version: V12.1

The AI Training feature in Aspen Plus enables you to use data and machine learning techniques to improve the accuracy of one or more models; the primary advantage of this approach over traditional data-driven machine learning models is that the training is done within the constraints of consistent physical properties and material and energy balances.

Machine learning is used only to compensate for deficiencies in the model. A neural network embedded within the model is trained using data to predict one or more model parameters as a function of feed and operating conditions.

Every exposed parameter in Aspen Plus can be replaced by a trained neural network.

Examples of such parameters include:

- Efficiencies in distillation columns
- Rate constants or reaction rates
- Heat transfer coefficients or heat transfer rates
- Mass transfer coefficients or mass transfer rates
- Drying rates
- Pipe friction factors
- Pump and compressor efficiencies

In this document, we highlight the different steps required to create such a First Principles Driven Hybrid Model to accurately capture these effects.

- Step 1 – Import Data
- Step 2 – Condition Data
- Step 3 – Evaluate Model
- Step 4 – Build, Train and Validate Model
- Step 5 – Deploy Model

General Workflow



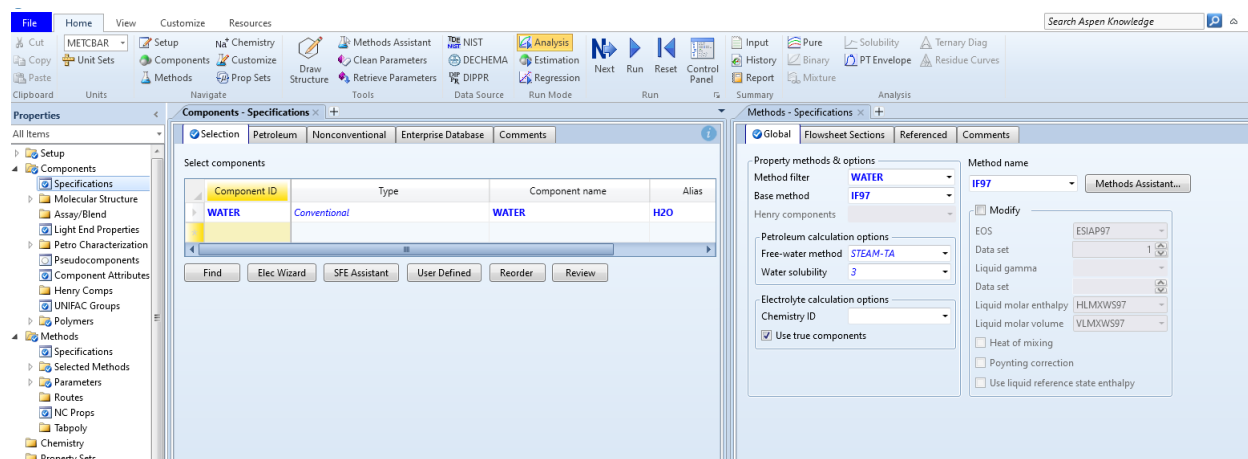
Heat Exchanger Heat Transfer Coefficient (UA).



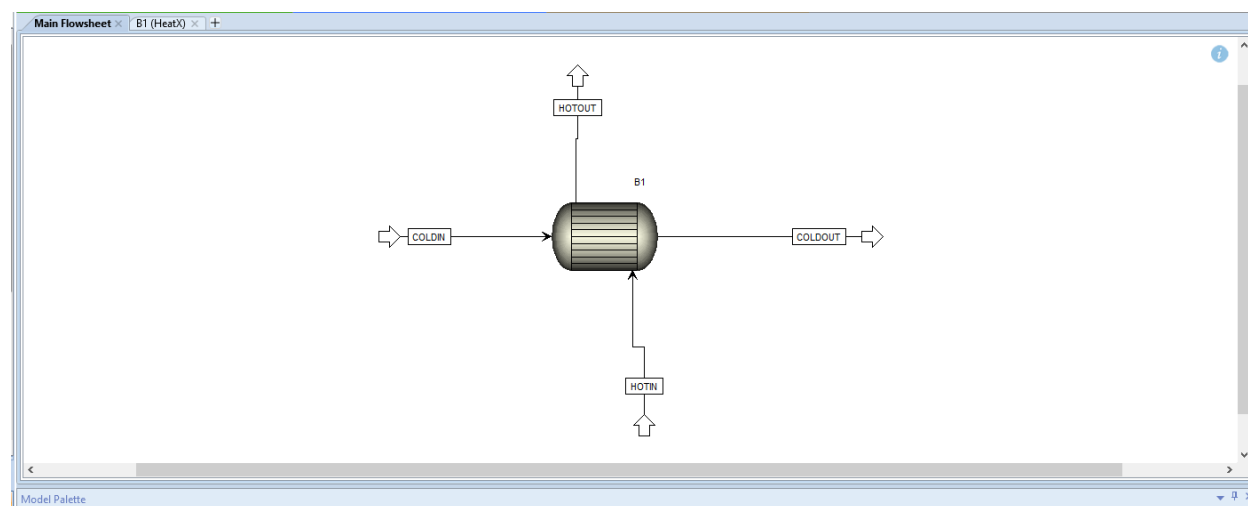
The objective of this example is to 'train' a model with plant data using First Principles-Driven Hybrid Models in Aspen Plus to help us optimize the performance of a Heat Exchanger's Heat Transfer coefficient (UA).

1. Start by opening **HeatXStarter.bkp**

The starter file has been defined for components with WATER only. IAPWS-IF97 is the selected property package.



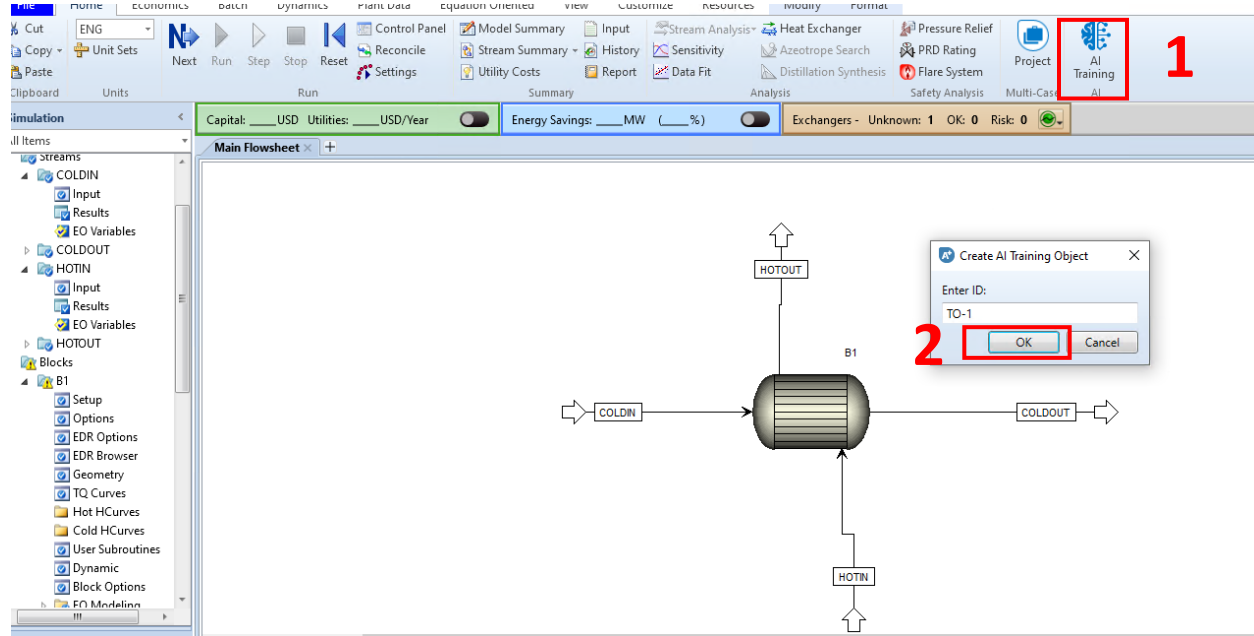
The simulation environment has already been defined and sets a HeatX block with its respective inlet and outlet streams on both hot and cold side, and a constant UA value of 8.5 BTU/hr-sft-R



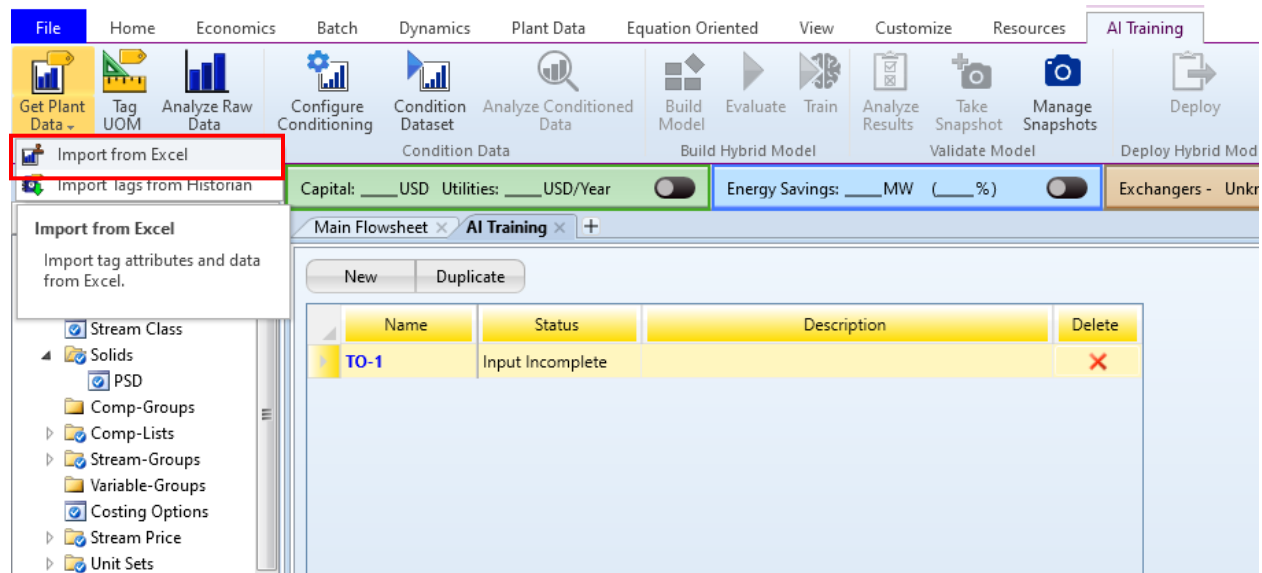
To train the model follow the listed steps:

Step 1- Import Data.

2. In the simulation Environment, click the **AI Training** icon, located at **Home tab | AI**. Click **OK** for the **AI Training Object** manager; the default name for our model TO-1.



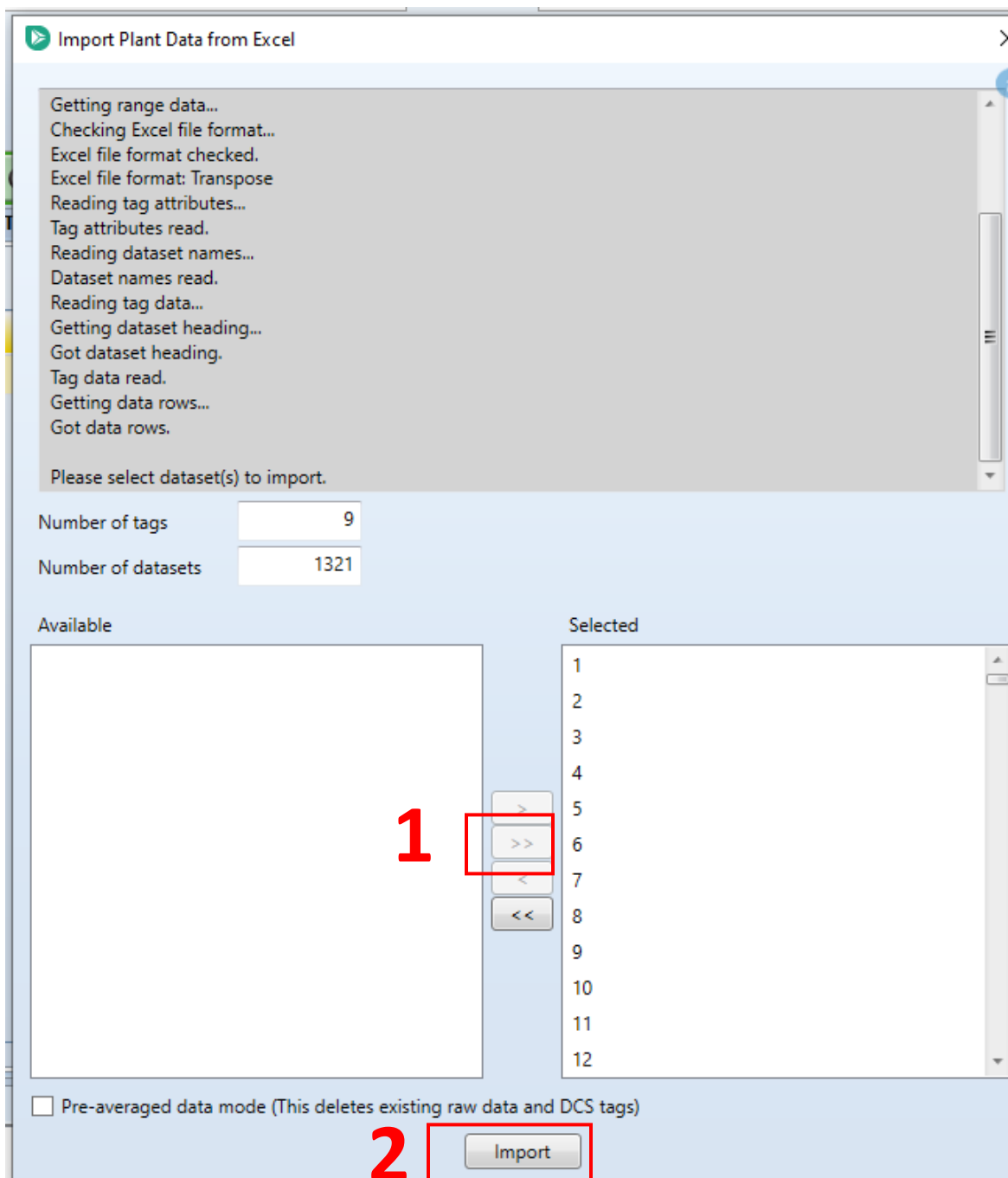
3. The AI Training tab will enable. Click the **Get Plant Data** button and from the options select **import from excel**, browse for the **heatx_PlantData_historian.xlsx** file and click open.





Note: Remember that the organization of this tab starts on the left side uploading the data and finishes with the deployment of your AI model at the right end.

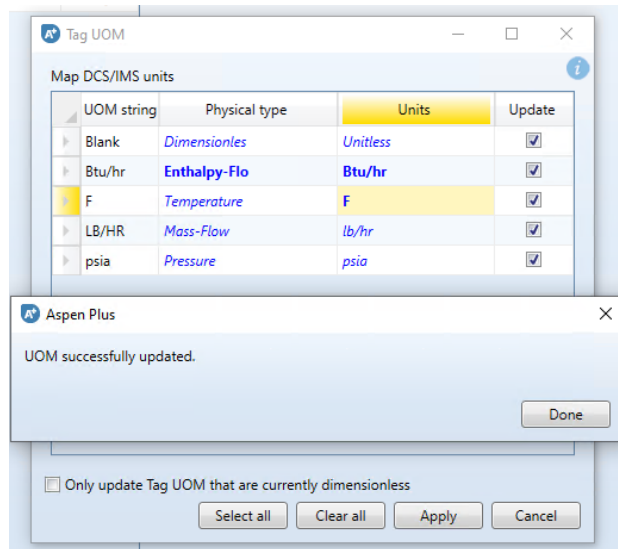
4. The **Import Plant Data manager** will display, and information will be loaded, it will show 9 tags and 1321 data sets. Select all of them, click the >> button to transfer all 1321 datasets; then click **Import**.





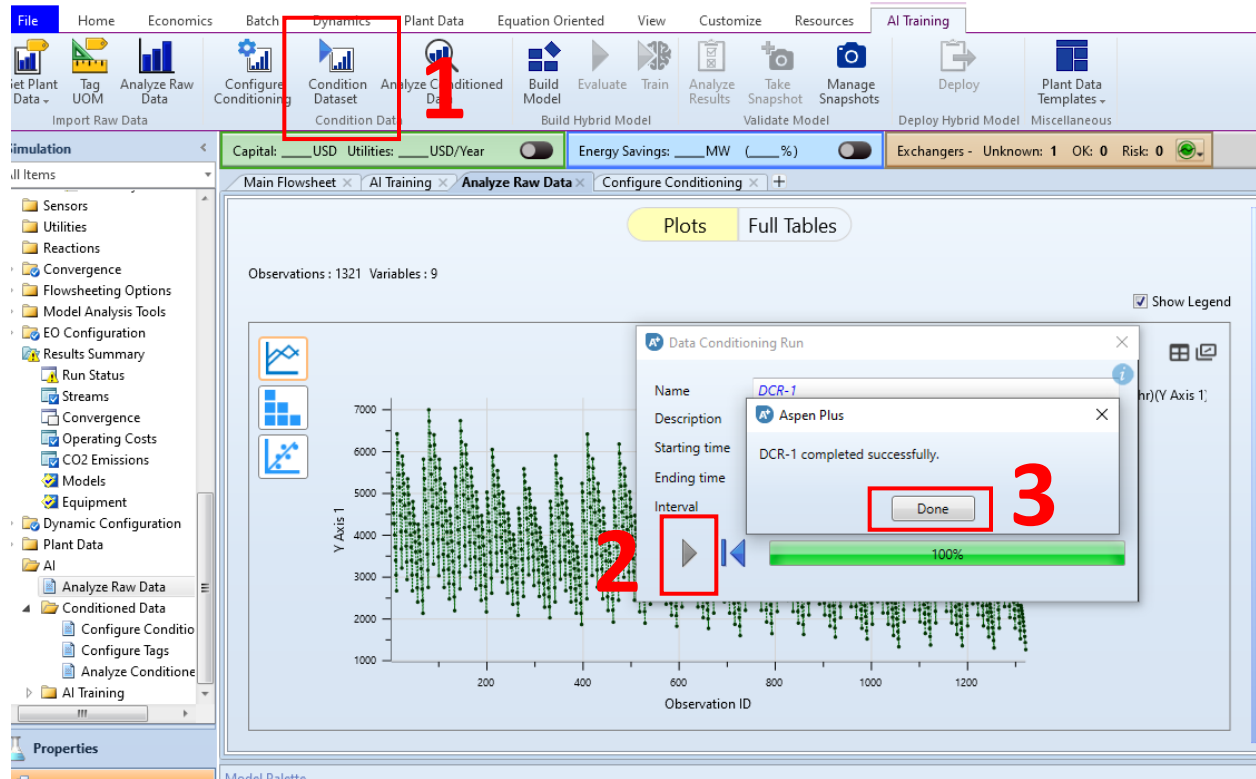
Note: once all data is imported the Import manager will display the message **Plant Data Successfully Imported**.

5. Close the Import Plant Data from Excel, and Map the DCS/IMS units, use **BTU/hr** for Enthalpy Flow, **F** for temperature, **lb/hr** for Mass Flow and **psia** for Pressure, click **Select all** then **Apply**.



Step 2- Conditioning Data.

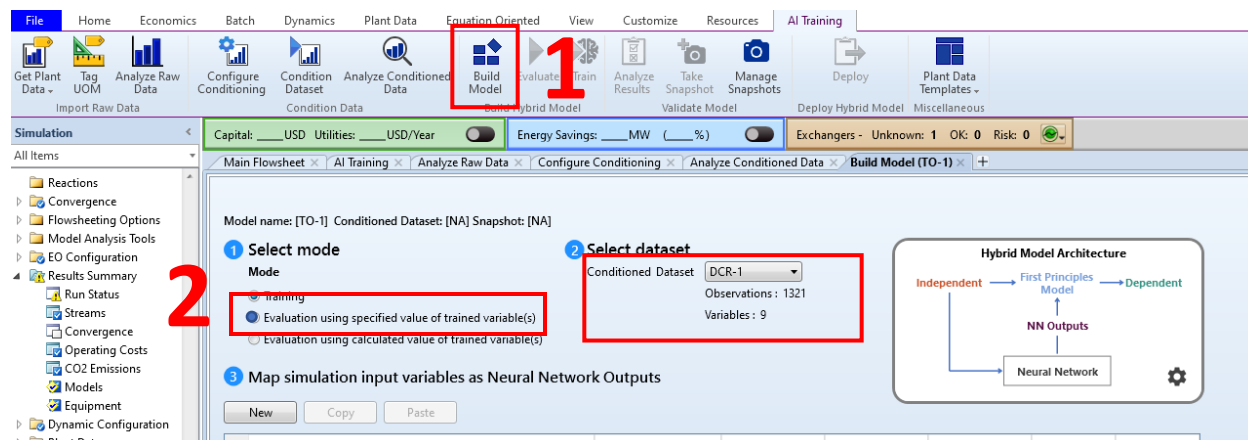
6. **Configure Conditioning Dataset** will not be needed for this case. Click the **Condition dataset**; the **Data Conditioning Run manager** will appear, click **Run** then **Done**



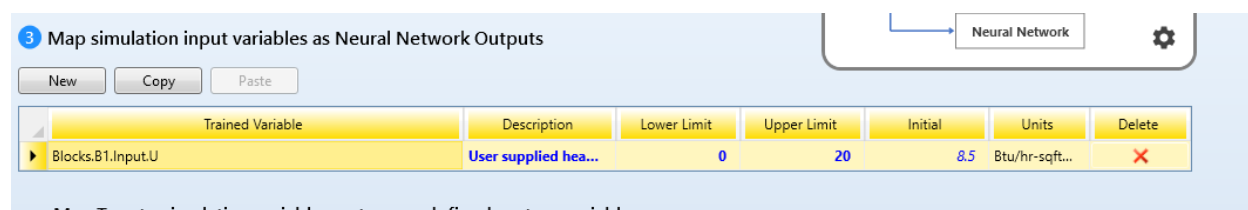
Note: Remember that if you detect any outlier, missing data, bad or oscillating data points, in **Configure Conditioning** you can define rules for the tags and data sets and set general conditions.

Step 3- Evaluate Model.

- Click the **Build Model** button at the AI Training ribbon and select **Evaluate Using Specified Value of Trained Variable(s)** under **Mode**, DCR-1 will be set as the conditioned set with 1321 observations and 9 variables.



8. Now map the simulation input variables. For input variables as Neural Network Outputs you can browse or copy and paste the variable **Heat Transfer Coefficient U** with a **lower limit** of 0, an **upper limit** of 20 and an **initial value** of 8.5, input for the Heat Exchanger Block.



9. Map the tags as independent or dependent variables, you can browse for them or copy and paste as follows:

Description	Stream/UnitOp.	Variable Name	Dependent/Independent
Mass flow hot feed	HOTIN	Mass Flow	Independent
Temperature hot feed	HOTIN	Temperature	Independent
Pressure hot feed	HOTIN	Pressure	Independent
Mass cold feed	COLDIN	Mass Flow	Independent
Temperature cold feed	COLDIN	Temperature	Independent
Pressure cold feed	COLDIN	Pressure	Independent
Duty	B1	Exchanger Cold Duty	Dependent
Temperature hot out	HOTOUT	Temperature	Dependent
Temperature cold out	COLDOUT	Temperature	Dependent



- 4 Map Tags to simulation variables or to user-defined custom variables
Assign each Tag as Independent or Dependent (simulation inputs = Independent, simulation outputs = Dependent)

☒ Include in Model	Tag	Description	Units	Variable Type	Variable Name	Independent	Dependent
☒	001:A1005.PV	MASS FLOW HOT INLET	lb/hr	Simulation	Streams.HOTIN2.Input.TOTFLOW.MIXED	☒	☐
☒	001:A1006.PV	temperature hot feed	F	Simulation	Streams.HOTIN2.Input.TEMP.MIXED	☒	☐
☒	001:A1007.PV	PRESSURE hot IN	psia	Simulation	Streams.HOTIN2.Input.PRES.MIXED	☒	☐
☒	001:A1008.PV	mass flow cold in	lb/hr	Simulation	Streams.COLDIN2.Input.TOTFLOW.MIXED	☒	☐
☒	001:A1009.PV	temperature cold feed	F	Simulation	Streams.COLDIN2.Input.TEMP.MIXED	☒	☐
☒	001:A1010.PV	PRESS cold in	psia	Simulation	Streams.COLDIN2.Input.PRES.MIXED	☒	☐
☒	001:A1011.PV	duty	Btu/hr	Simulation	Blocks.B12.Output.HX_DUTY	☐	☒
☒	001:A1012.PV	temperature hot out	F	Simulation	Streams.HOTOUT2.Output.STR_MAIN.TEMP	☐	☒
☒	001:A1013.PV	temperature cold out	F	Simulation	Streams.COLDOUT2.Output.STR_MAIN.TEMP	☐	☒

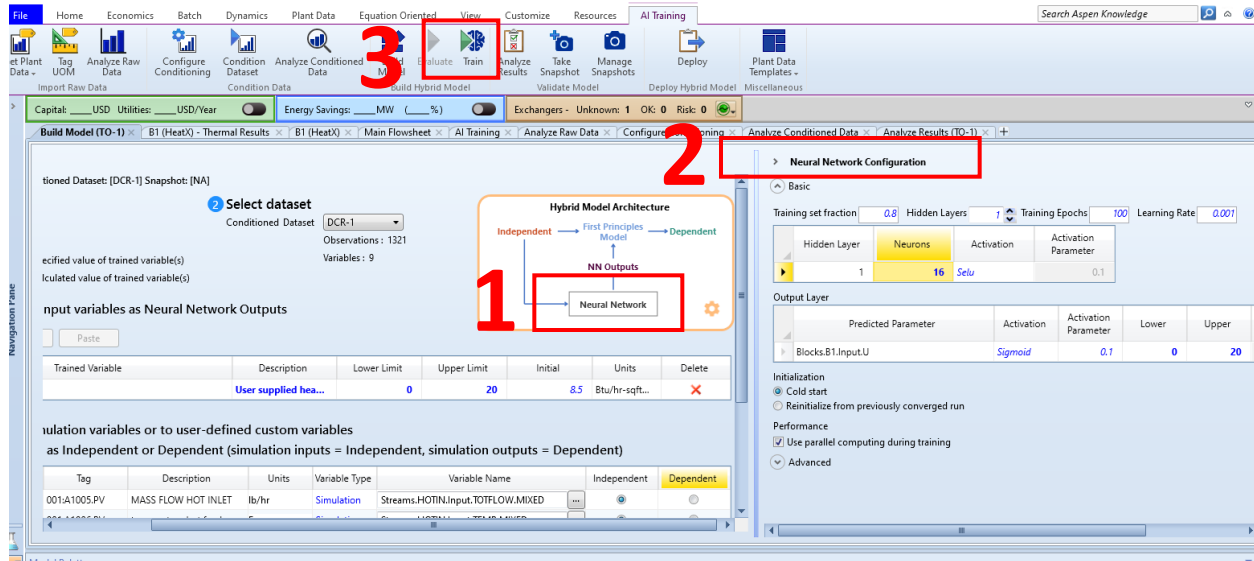
10. Click Evaluate and Observe the results.

Step 4 – Build, Train and Validate model.

11. Go back to **Build Model** and select **Training** mode.

The screenshot shows the Aspen Tech software interface with the 'Build Model' tab selected. The 'Mode' dropdown is set to 'Training'. The 'Select dataset' section shows 'Conditioned Dataset' as 'DCR-1' with 'Observations: 1321' and 'Variables: 9'. The 'Hybrid Model Architecture' diagram shows 'Independent' and 'Dependent' variables connected by a 'Neural Network' block.

12. Click **Neural Network** button under Hybrid Model Architecture manager and for **Training set fraction** enter 0.8, **Hidden layers** 1, **Training Epochs** 30. 16 **Neurons** and select Sigmoid for the **activation function**. Click Train.



13. Results for the AI will display showing approximately 0.99 accuracy in the Dependent Variables Fit Map plots with the relationships of variables and the Duty and outlet temperatures for both hot and cold sides.



Step 5 – Deploy Model.

14. Model is ready to be deployed, click **Deploy**, TO-1 AI model will not be attached to the HeatX Block, with the trained variables.

